

Manas Pandey

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Summary

Multidisciplinary Engineer with published research in **RF/SDR** and deep expertise in **Systems Programming** and **AI/ML**.

Passionate about **building scalable systems** and seeking to **solve complex challenges** by leveraging unique hardware & software insights.

Education

- Manipal University Jaipur** Jaipur, India
Bachelor of Technology - Electronics and Communications 2021 - 2025
Courses: Computer Organization & Architecture, Embedded Systems, Microprocessors & Microcontrollers, Digital Signal Processing, Analog & Digital Communication, Signals & Systems, Electromagnetic Field Theory, Control Systems, Circuits & Networks

Skills

- Languages:** Python, SQL, C, C++, JavaScript, MATLAB
- Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face
- Tools:** Docker, Git/GitHub, Jupyter, Apache (Airflow, Spark), RESTful APIs, MySQL, PostgreSQL, LaTeX, MQTT
- Platforms & Cloud:** AWS (SageMaker, S3, Lambda, EC2, IoT Core, CloudWatch), Linux, Raspberry Pi, GNU Radio
- Soft Skills:** Problem Solving, Algorithm Design & Optimization, Documentation, R&D, Stakeholder Communication
- Concepts & Methodologies:** MLOps, LLMops, Fine-tuning, Transformer Architectures, Deep Learning, NLP, Computer Vision, Data Pipelines, Transfer Learning, Reinforcement Learning, Supervised/Unsupervised Learning, CI/CD, Containerization, Serverless Architecture, Cloud Computing, OOP, Data Structures & Algorithms, Network Protocols

Projects

- GitLite (Distributed VCS, Systems Programming, Git Protocol, Zlib Compression, SHA-1) [Repo] [Article]:** Built fully functional distributed version control system from scratch: init, add, commit, log, branch, checkout, tree, blob, zlib compression, exact Git object format. Achieved 100% perfect object interoperability with multi-commit repositories.
Tech: C++, Zlib, POSIX, OpenSSL (SHA-1)
- DBLite (In-Memory Database, Concurrency, Persistence) [Repo] [Article]:** Implemented production-grade Redis clone supporting strings, lists, hashes, sets, sorted sets, TTL expiry, RDB/AOF persistence, and gevent threading. Passes full Redis test suite on port 6379 with zero dependencies.
Tech: Python, Gevent, Collections.deque, Socket.
- HTTP Server (Event-Driven Web Server, WSGI, Asynchronous I/O, HTTPS) [Repo] [Article]:** Developed single-threaded asynchronous HTTP/1.1 + WSGI server using selectors, supporting Keep-Alive, chunked encoding, SSL/TLS, and 500+ concurrent connections with zero dependencies.
Tech: Python, Selectors, Socket, SSL.
- C-Like (Compiler Design, Recursive Descent Parsing, Semantic Analysis) [Repo] [Article]:** Built full interpreter from scratch with lexer, parser, AST, multi-scope symbol tables, recursion, arrays, and function calls. Successfully executes complex recursive programs and test suites with zero dependencies.
Tech: Python, AST, Visitor Pattern
- BareCodeX (Video Codec, YUV420, Delta Encoding, Multimedia Compression) [Repo] [Article]:** Engineered a full video encoder/decoder from scratch, performing RGB → YUV420 subsampling, inter-frame delta encoding, run-length encoding, and DEFLATE compression, achieving 80–95% file size reduction on raw video while preserving visual quality.
Tech: C++, Zlib, Raw Pixel Processing.
- Flexible Microstrip Patch Antenna (Bendable Design for Wearables/IoT) [Article]:** Designed & simulated ultra compact flexible antenna on PET substrate (20×30 mm) achieving -26.5 dB S11, VSWR ≈ 1.1, and 100-200 MHz bandwidth at 1.896 GHz resonance. Engineered for conformal/wearable applications with minimal performance shift under bending.
Tech: CST Studio Suite, CAD, 3D modeling.

Experience

- Manipal University Jaipur - Communications Laboratory** Jaipur, India
Researcher March 2022 - March 2023
 - Multi-Tone Spectrum Sensing Testbed:** Built real-time GNU Radio + Amitec SDR testbed detecting simultaneous PU signals at 20/45/70/100 kHz with 100% detection accuracy and zero false alarms in noise-only cases.
 - High-Resolution Energy Detection:** Implemented 1024-point FFT flowgraph (1 Msps, 976 Hz RBW) achieving < 1% frequency estimation error and clear H0/H1 decision separation across all tones.
 - Threshold Optimization & Noise Resilience:** Optimized fixed energy threshold ($\lambda \approx 12-13$) to eliminate false alarms while maintaining $P_d = 1.0$ in controlled SNR, enabling robust weak-signal detection.
 - Experimental Validation & Statistical Analysis:** Validated system with 50+ over-the-air experiments using dual SDRs; achieved negligible processing delay (< 100 ms) and perfect busy-channel identification.

Publications

- Experimental Investigation on Spectrum Sensing Testbed Using GNU Radio and SDR:** Published in Springer Nature on 28 July 2023 in ICCDC 2023, Springer [DOI: 10.1007/978-981-99-2710-4_24].

Certifications

- Courses:** IBM Generative AI Engineering [IBM], IBM Deep Learning with PyTorch/Keras/TensorFlow [IBM], Power Electronics [University of Colorado Boulder], MLOps [Duke University], LLMops [Duke University], IBM Data Science [IBM], AWS Cloud Solutions Architect [AWS]